

St. Edward's University

The Bill Munday School of Business

ECON 3335: Urban and Regional Economics
Spring 2027

Instructor: Chen Xu
Office: Trustee Room 309
Email: cxu@stedwards.edu
Office Hours: TBD
Class Location: TBD
Class Times: TBD
Exam Location: TBD
Final Exam Time: TBD

Please take time to watch this video

The Bill Munday School of Business Mission Statement

The Bill Munday School of Business delivers an experiential, values-based education anchored in the liberal arts and the Holy Cross traditions of learning and service. The School educates students to excel professionally in the dynamic 21st century business environment by emphasizing creative thinking, critical analysis, global collaboration, lifelong learning, adaptation to change, and ethical behavior. The School promotes relationships on our campus, with our global partners, and across the rich ecosystem of Austin, focusing on technology, creativity, social enterprise, and emerging trends.

The Bill Munday School of Business Policy on Ethics: Although topics in this class will be examined mainly from an economic way of thinking, the ethical components of these issues will be raised and discussed whenever relevant.

Course Description

This course will introduce you to core issues and recent advances in urban economics. The first part develops canonical models of urban economics to explain spatial regularities within and across cities, and presents the market forces that cause cities to grow. Then, topics such as housing, neighborhood effects, congestion, local government, segregation, violence, and disease are analyzed. Along with classes, lab sessions expose you to practical tools in data science such as data cleaning, visualization, and mapping with the use of programming language. By the end of

the semester, the student will have a good understanding of the economic approach to cities and will be comfortable using R.

Student Learning Outcomes

The goal of the course is enable student to understand the following:

1. Understand why cities exist, what drives difference in success across cities.
2. Articulate the efficacy of various placed-based policies.
3. Demonstrate a thorough understanding of the urban economics model.

Prerequisites

- Econ 2301.
- Economics concepts are often presented in mathematical models. The course assumes proficiency in algebra and geometry. Calculus is not required.

Textbook and other materials

This course does not follow one textbook.

1. Brueckner, J.K.(2011) *Lectures on Urban Economics*, MIT Press.
2. Glaeser, E. (2011). *Triumph of the City*. Penguin Books.
3. O'Sullivan, A. (2007). *Urban Economics, 9th Edition*, by Arthur O'Sullivan McGraw Hill Irwin, ISBN: 9780262016360.

Canvas and SEU Email

- Class announcements and homework will be posted on the course website on Canvas: <https://canvas.stedwards.edu>. It is your responsibility to check the site regularly.

Grading Standards

Attendance	5%
Homework (5 assignments)	35%
Midterm Exam	20%
Lab Quiz	20%
Final Group Project	20%

Your grade will be determined by the following criteria:

- Attendance (5%)

- Attendance is required for this course. However, everyone is allowed to have up to 3 absences. You will not receive full credit on attendance if you are absent for more than 3 classes.
- Homework (35%)
 - Homework should be completed and submitted online through Moodle. Late assignments will receive a daily penalty of 5%, and you can submit your late assignment up to 3 days after the original deadline. After that, assignments will receive no credit. You are required to submit two files: .rmd and .pdf. Points might be taken due to lack of organization in your report (no more than 5%).

Students are encouraged to work together on homework. However, sharing, copying, or providing any part of a homework solution or code is an infraction of the University's rules on Academic Integrity. Any violation will be punished as severely as possible.
- Pop Quizzes (15%)
 - There will be several quizzes throughout the semester. Dates for the quizzes will be announced ahead. The goal of giving the quizzes is to help you understand the materials.
- Final Project (20%)
 - There will be a Final Group Project. Details TBA.

Grading scale

Both exams will be out of 100 points. At the end of the course, I will compute a final percentage grade based on component percentages of each grade category using the weights given above. I will then convert this final percentage grade into letter grades according to the following scale (where g indicates your final percentage grade):

$98\% \leq g \leq 100\%$	A+
$92\% \leq g \leq 97.99\%$	A
$90\% \leq g \leq 91.99\%$	A-
$88\% \leq g < 89.99\%$	B+
$82\% \leq g \leq 87.99\%$	B
$80\% \leq g \leq 81.99\%$	B-
$77\% \leq g < 79.99\%$	C+
$70\% \leq g < 76.99\%$	C
$60\% \leq g < 70\%$	D
$0\% \leq g < 60\%$	F

I reserve the right to scale upwards everyone's final percentage grades.

Policy on use of Artificial Intelligence (AI)

Generative AI technologies (GAIs) are any artificial intelligence technology that can produce or summarize digital content (e.g., text, images, video, audio) from a prompt. Current examples include ChatGPT, Bard, Perplexity, the Google Docs "Help me write" feature, Dall-E, and Midjourney.

Any work written, developed, created, or inspired by artificial intelligence (AI) is considered plagiarism and will not be tolerated. While the ever-changing new developments with AI will find their place in our workforces, personal lives, and even in discussions in the classroom, in the realm of education and learning, this kind of technology does not work and does not belong. Simply put, college is a place for learning and a place for learning how to problem-solve, not use AI to solve the problem.

- This policy applies to all of your work in the classes that I teach.
- I don't make policy for other classes. You will have to ask your other professors about their policies.
- Because AIs are fast-emerging and are evolving constantly, I may revise this policy at any time.

Drop/Withdrawal Policy

Discontinuing attendance or notifying an instructor of a status change does not constitute a drop or withdrawal. The student is responsible for initiating and completing the drop or withdrawal procedure by utilizing myHilltop. If the student fails to complete the process, s/he will receive the grade(s) assigned by the instructor(s) of record and be billed accordingly.

A student may add or drop courses (make schedule adjustments) anytime during the registration period. After this period, a student may only withdraw from courses. Consult the Registrar Dates & Deadlines and Course Schedules online for specific course begin, end, add, drop or withdrawal dates and other important deadlines:

<https://www.stedwards.edu/academics/registrar/academic-calendar/dates-deadlines>

A student may withdraw from a course under the following provisions:

1. Through the 12th class day (Official Headcount Day) of any semester (or the equivalent during short terms), without having the course recorded on the transcript.
2. After the 12th class day of the semester (or the equivalent during short terms), but before the end of the 10th week of the semester, with a grade of W recorded on the transcript. **If the withdrawal procedure is not followed to its conclusion, a grade of F will be entered on the transcript.**
3. After the first 10 weeks of a semester, four weeks of a six-week term, or eight weeks of a 12-week term, no student may withdraw from a course.

During the first 10 weeks of a semesters and the comparable period in short terms, an instructor has the option, but is not required, to request the Registrar to assign a grade of WA

(*Withdrawal due to Absences*) to a student who has excessive absences from a class.

To reiterate: A student is not officially dropped from a class, except in the case of a WA, until the student completes the drop process, as published by the Office of the Registrar.

For more information, consult your Student Bulletin, found here:

<https://www.stedwards.edu/policies/university-handbooks-guides-and-manuals>.

Incomplete

A grade of I (Incomplete) may be given only in exceptional cases in which extenuating circumstances that occur in the last 10 days of the semester interfere with the completion of final course work. Work must be completed and the grade received in the Office of the Registrar by Oct. 1 for all summer-term courses, March 1 for fall semester courses and July 1 for spring semester courses, or a grade of F will be entered on the transcript.

University Policies

Academic Integrity and Ethical Components

St. Edward's University expects academic honesty from all members of the community, and it is our policy that academic integrity be fostered to the highest degree possible. The academic integrity policy will be strictly enforced in this class.

An academic community of integrity strives for excellence by creating a learning environment where high academic standards in terms of student performance, program integrity, course development and requirements are encouraged. This can only be realized if the community is aware of and adheres to clearly established values and goals where the end is personal and academic integrity. Our commitment to the values of academic integrity contributes to St. Edward's University's being a quality institution of higher learning and that adherence to the Mission Statement will continue to be fulfilled.

The ability to express ourselves articulately in both oral and written form requires ability for critical and creative thinking which necessitates a clear sense of academic integrity. Academic integrity is important because it involves values which can be grounded in both reason and faith. The core values of academic and personal integrity are:

1. **Honesty:** An academic community of integrity advances the quest for truth and knowledge by requiring intellectual and personal honesty in learning, teaching, research, and service.
2. **Fairness:** An academic community of integrity establishes clear standards, practices, and procedures and expects fairness in the interactions of students, faculty, and administrators.
3. **Trust:** An academic community of integrity fosters a climate of mutual trust, encourages the free exchange of ideas, and enables all to reach their highest potential.
4. **Respect:** An academic community of integrity recognizes the participatory nature of the learning process and honors and respects the work of others and the members of the learning community.

5. Responsibility: An academic community of integrity upholds personal accountability and depends upon action in the face of wrongdoing.

We achieve this not only through publication and proclamation of these values, but also by creating a learning environment where these values can be realized. St. Edward's University expects academic honesty from all members of the community, and it is our policy that academic integrity be fostered to the highest degree possible. Consequently, all work submitted for grading in a course must be created as a result of your own thought and effort. Representing work as your own when it is not a result of much thought and effort is a violation of our code of academic integrity. Whenever it is established that academic dishonesty has occurred, the course instructor shall impose a penalty upon the offending individual(s). It is recognized that some offenses are more egregious than others and that, therefore, a range of penalties should be available. Whenever possible, it would also be important to try to determine the intent of the offender, since the error could be a result of careless work rather than intent to deceive. The maximum penalty for a first offense is failure in the course, and if that penalty is imposed, the student does not have the option of withdrawing from the course. In cases of mitigating circumstance, the instructor has the option of assigning a lesser penalty. After obtaining sufficient evidence that such dishonesty has occurred, the instructor should discuss the question with the student. Instructors who impose a penalty for serious academic dishonesty should report this penalty to the dean or director of the program in which the course is offered. The dean or director will report confirmed cases of dishonesty to the Associate VP for Academic Affairs, and a record of all offenses will be kept in the Office of the Associate VP for Academic Affairs. Offenses that resulted in failure in the course will then be reported to the Office of the Registrar.

The Associate VP for Academic Affairs will determine whether an earlier serious offense by the student has been recorded. A second serious offense merits an automatic appeal. Upon denial of this appeal, the Associate VP for Academic Affairs will dismiss the student from the university and notify the registrar and the dean or director.

A procedure for student appeal is already established and is outlined in detail in the Student Handbook, but it should be noted that, for appeals of decisions regarding academic dishonesty, the student must appeal to the dean of the school or program director within five (5) working days after being notified by the instructor.

For more information, visit: <https://stedwards.app.box.com/v/academicbulletins>.

Student Disability Services

Student Disability Services coordinates reasonable accommodations for students with documented disabilities (medical, learning or psychological). Any student who feels s/he may need an accommodation based on the impact of a disability should follow the university's accommodation procedure by contacting Student Disability Services (512-448-8561 or Moody Hall 155).

Title IX Mandatory Reporting

A Note on Sex/Gender Harassment, Discrimination & Misconduct

St. Edward's University is committed to fostering a safe, inclusive and productive learning environment. Title IX, state laws, and SEU policy prohibit discrimination on the basis of sex and gender identity. Consequently, sexual misconduct—including harassment, relationship violence,

sexual assault, and stalking —is also prohibited at SEU.

The university encourages anyone impacted by or experiencing sexual misconduct to talk to someone about what happened, so they can get the support they need and SEU can respond appropriately.

But you should know that only some SEU employees can offer you confidentiality. I cannot.

If you wish to speak confidentially about an incident of sexual misconduct, please contact the Health & Counseling Center in Johnson Hall at 512-448-8538 to speak with a counselor or advocate.

Please know as a faculty member, I am a mandatory reporter and required by SEU, federal and state laws to report incidents of sexual misconduct that I "witness" or "receive information regarding the occurrence of " and thus cannot offer any confidentiality —even if you ask me to. I must provide our Title IX coordinator with relevant details including the names of those involved in the incident or I will be terminated from employment and possibly prosecuted.

To make a formal report, you can contact the Dean of Students Office in Main Building, G 16, 512-448-8408, or go to <https://www.stedwards.edu/campus-services/title-ix-and-preventing-discrimination/title-ix-students>. Anonymous reporting is an option through the website. If you would like to make a police report, you can contact our University Police Department at 512-448-8444. The university will investigate reports of sexual misconduct and may need to override a request for confidentiality and pursue an alleged perpetrator in order to provide a safe campus for everyone.

If you have questions about university policies and procedures regarding sexual misconduct, please contact our Title IX Coordinator, Dr. Lisa Kirkpatrick, Vice President for Student Affairs, Main Building, G 16, 512-448-8425. You can also learn more about Title IX at St. Edward's University by visiting the website at <https://www.stedwards.edu/campus-services/title-ix-and-preventing-discrimination>.

Tentative Course Schedule

Class readings are subject to change, contingent on mitigating circumstances and the progress we make as a class. Students are encouraged to attend all lectures and check the course website for updates.

Week 01, 01/26–01/28: Syllabus Day and Intro to R & Stylized Facts of Cities

- Intro to Econ 414 – Urban Economics
- Using Rmarkdown
- Urban regularities

Week 02, 02/02–02/04: Market Forces in the Development of Cities & Dataviz with R
Readings: Moretti (2012) ch. 4, O'Sullivan (2007) ch. 2 and 3

- Why do cities exist?

- Why do firms cluster?
- Data cleaning and visualization with Tidyverse

Week 03, 02/09–02/11: Monocentric City Model I

Reading: Brueckner (2011) ch. 2

- The Alonso-Muth-Mills Model

Week 04, 02/16–02/18: Monocentric City Model II & Sprawl and Zoning *Readings:* Brueckner (2011) ch. 3 and 4

- The Alonso-Muth-Mills Model: Extensions
- Urban sprawl and land-use controls
- **Homework 1 due 02/15**

Week 05, 02/23–02/25: Urban Growth and Decline & The Great Divergence

Readings: Glaeser (2011) ch. 2 & 9, Moretti (2012) ch. 3

- Why do cities decline?
- How do cities succeed?
- The Great Divergence

Week 06, 03/02–03/04: Spatial Equilibrium Across Cities & Mapping Data

- Rosen-Roback Model
- Mapping data with tmap and Leaflet
- **Homework 2 due 03/01**

Week 07, 03/09–03/11: Midterm Week

- **Midterm Exam 03/11**

Week 08, 03/16–03/18: Urban Transportation

Reading: Brueckner (2011) ch. 5

- Urban transit, traffic allocations
- Congestion Pricing
- **Homework 3 due 03/15**

Week 09, 03/23–03/25: Housing & Housing Policies

Readings: Brueckner (2011) ch. 6 and 7

- Hedonic Model & Tenure Choice
- Housing Policies & Affordable Housing

Week 10, 03/30–04/01: Local Government

Reading: Brueckner (2011) ch. 8

- The Tiebout Model
- **Homework 4 due 03/29**

Week 11, 04/06–04/08: Lab Quiz Week

- **Lab Quiz 04/08**

Week 12, 04/13–04/15: Neighborhood Choice

Readings: O’Sullivan (2007) ch. 8, Glaeser (2011) ch. 3

- April 13 Break
- Segregation trends
- Tipping model of neighborhoods
- Rise and fall of the American ghetto
- Consequences of segregation

Week 13, 04/20–04/22: Neighborhood Effects *Readings:* Lecture notes based on Sampson (2012)

- “[...] near things are more related than distant things”
- Moving to Opportunity Experiment

Week 14, 04/27–04/29: Exploratory Spatial Data Analysis (ESDA)

- Modelling proximity with spatial weight matrices
- Spatial autocorrelation, Moran’s I, and LISA map
- Spatial analysis using R
- **Homework 5 due 04/26**

Week 15, 05/04–05/06: Violence & Disease

- The criminology of place
- 1854 Broad Street cholera outbreak
- COVID-19 Spatial Distribution
- Final Group Project Discussion
- May 6 Reading Day

Week 16, 05/11–05/13: Final Project Week

- **Final Group Project due 05/14**